

# Linear Transformations

## 7.1 Linear Transformations: Definition and Examples

### Definition of a Linear Transformation

A **linear transformation** is a function  $T: V \rightarrow W$  between two vector spaces  $V$  and  $W$  that satisfies the following two conditions for all vectors  $\mathbf{u}, \mathbf{v} \in V$  and all scalars  $c \in \mathbb{R}$ :

1. **Additivity:**  $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ .
2. **Homogeneity:**  $T(c\mathbf{u}) = cT(\mathbf{u})$ .

In essence, a linear transformation preserves vector addition and scalar multiplication.

### Examples of Linear Transformations

1. **Zero Transformation:**  $T: V \rightarrow W$  defined by  $T(\mathbf{v}) = \mathbf{0}$  for all  $\mathbf{v} \in V$ .
2. **Identity Transformation:**  $T: V \rightarrow V$  defined by  $T(\mathbf{v}) = \mathbf{v}$  for all  $\mathbf{v} \in V$ .
3. **Scalar Multiplication Transformation:**  $T: V \rightarrow V$  defined by  $T(\mathbf{v}) = c\mathbf{v}$  for some scalar  $c \in \mathbb{R}$  and all  $\mathbf{v} \in V$ . This is a scaling operation.
4. **Matrix Transformation:**  $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$  defined by  $T(\mathbf{x}) = A\mathbf{x}$ , where  $A$  is a fixed  $m \times n$  matrix and  $\mathbf{x} \in \mathbb{R}^n$  is a column vector.
5. **Projection Transformation:**  $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$  that projects each vector onto a fixed subspace  $W$  of  $\mathbb{R}^n$ .
6. **Differentiation:**  $T: P_n \rightarrow P_{n-1}$  defined by  $T(p(x)) = p'(x)$ , where  $P_n$  is the vector space of polynomials of degree at most  $n$ .
7. **Integration:**  $T: C[a, b] \rightarrow \mathbb{R}$  defined by  $T(f) = \int_a^b f(x) dx$ , where  $C[a, b]$  is the vector space of continuous functions on the interval  $[a, b]$ .

### Non-Examples of Linear Transformations

1.  $T(\mathbf{x}) = \mathbf{x} + \mathbf{b}$  where  $\mathbf{b} \neq \mathbf{0}$ . This transformation is not linear because  $T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$ , violating the linearity condition.
2.  $T(x) = x^2$ . Does not respect scalar multiplication, e.g.  $T(2x) = 4x^2$  while  $2T(x) = 2x^2$ .

### Properties of Linear Transformations

If  $T: V \rightarrow W$  is a linear transformation, then:

1.  $T(\mathbf{0}_V) = \mathbf{0}_W$ , where  $\mathbf{0}_V$  and  $\mathbf{0}_W$  are the zero vectors in  $V$  and  $W$ , respectively.
2.  $T(-\mathbf{v}) = -T(\mathbf{v})$  for all  $\mathbf{v} \in V$ .
3.  $T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \dots + c_kT(\mathbf{v}_k)$  for all  $\mathbf{v}_i \in V$  and  $c_i \in \mathbb{R}$ . This means a linear transformation preserves linear combinations.

## 7.2 Kernel and Range of a Linear Transformation

### Kernel (Null Space)

The **kernel** (or null space) of a linear transformation  $T: V \rightarrow W$  is the set of all vectors in  $V$  that are mapped to the zero vector in  $W$ .

$$\ker(T) = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W\}.$$

The kernel is a subspace of  $V$ .

### Range (Image)

The **range** (or image) of a linear transformation  $T: V \rightarrow W$  is the set of all vectors in  $W$  that are images of vectors in  $V$  under  $T$ .

$$\text{range}(T) = \{\mathbf{w} \in W \mid \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}.$$

The range is a subspace of  $W$ .

## Example: Matrix Transformation

Let  $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$  be defined by  $T(\mathbf{x}) = A\mathbf{x}$ , where  $A$  is an  $m \times n$  matrix. Then:

- **Kernel:**  $\ker(T) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$ , which is the null space of the matrix  $A$ .
- **Range:**  $\text{range}(T) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\}$ , which is the column space of the matrix  $A$ .

## Finding the Kernel and Range

1. **Kernel:** To find the kernel of  $T$ , solve the equation  $T(\mathbf{v}) = \mathbf{0}_W$  for  $\mathbf{v}$ . The set of all solutions forms the kernel.
2. **Range:** To find the range of  $T$ , determine the set of all possible outputs of  $T$ . In the case of a matrix transformation, find a basis for the column space of the matrix.

## Theorem: Dimension Theorem (Rank-Nullity Theorem)

Let  $T: V \rightarrow W$  be a linear transformation, where  $V$  is a finite-dimensional vector space. Then:

$$\dim(V) = \dim(\ker(T)) + \dim(\text{range}(T)).$$

In the context of a matrix transformation  $T(\mathbf{x}) = A\mathbf{x}$ , this translates to:

$$(\text{number of columns of } A) = \text{nullity}(A) + \text{rank}(A).$$

## Definitions: Rank and Nullity

- The **rank** of a linear transformation  $T$  is the dimension of its range:  $\text{rank}(T) = \dim(\text{range}(T))$ .
- The **nullity** of a linear transformation  $T$  is the dimension of its kernel:  $\text{nullity}(T) = \dim(\ker(T))$ .

## 7.3 Matrix Representation of a Linear Transformation

### Coordinate Vectors

Let  $V$  be a finite-dimensional vector space with an ordered basis  $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ . The **coordinate vector** of a vector  $\mathbf{v} \in V$  with respect to the basis  $B$  is the column vector

$$[\mathbf{v}]_B = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix},$$

where  $c_1, c_2, \dots, c_n$  are the unique scalars such that  $\mathbf{v} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n$ .

### Matrix Representation

Let  $T: V \rightarrow W$  be a linear transformation, where  $V$  and  $W$  are finite-dimensional vector spaces with ordered bases  $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$  and  $C = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ , respectively. Then the **matrix representation** of  $T$  with respect to the bases  $B$  and  $C$  is the  $m \times n$  matrix  $A$  defined by

$$A = [[T(\mathbf{b}_1)]_C \quad [T(\mathbf{b}_2)]_C \quad \dots \quad [T(\mathbf{b}_n)]_C].$$

That is, the  $j$ -th column of  $A$  is the coordinate vector of  $T(\mathbf{b}_j)$  with respect to the basis  $C$ .

### Fundamental Property

The matrix representation  $A$  of  $T$  satisfies the fundamental property

$$[T(\mathbf{v})]_C = A[\mathbf{v}]_B$$

for all vectors  $\mathbf{v} \in V$ .

### Algorithm to Find the Matrix Representation

1. Choose ordered bases  $B$  and  $C$  for  $V$  and  $W$ , respectively.
2. Compute  $T(\mathbf{b}_j)$  for each basis vector  $\mathbf{b}_j \in B$ .
3. Find the coordinate vector  $[T(\mathbf{b}_j)]_C$  of each  $T(\mathbf{b}_j)$  with respect to the basis  $C$ .
4. Form the matrix  $A$  by using these coordinate vectors as columns.

## Change of Basis

Let  $V$  be a finite-dimensional vector space, and let  $B$  and  $C$  be two ordered bases for  $V$ . Then the **change-of-basis matrix** from  $B$  to  $C$  is the matrix  $P_{C \leftarrow B}$  defined by

$$P_{C \leftarrow B} = [[\mathbf{b}_1]_C \quad [\mathbf{b}_2]_C \quad \dots \quad [\mathbf{b}_n]_C].$$

For any vector  $\mathbf{v} \in V$ , we have

$$[\mathbf{v}]_C = P_{C \leftarrow B}[\mathbf{v}]_B.$$

## Theorem: Combining Linear Transformation With Change of Basis.

Let  $T: V \rightarrow W$  be a linear transformation, where  $V$  and  $W$  are finite-dimensional vector spaces.

Let  $B, B'$  be two ordered bases for  $V$  and  $C, C'$  be two ordered bases for  $W$ . Let  $A$  be the matrix representation of  $T$  with respect to the basis  $B$  and  $C$  and let  $A'$  be the matrix representation of  $T$  with respect to the basis  $B'$  and  $C'$ .

Let  $P$  be the change of basis matrix from  $B'$  to  $B$  and let  $Q$  be the change of basis matrix from  $C'$  to  $C$ .

Then we have:

$$A' = Q^{-1}AP$$

## 7.4 Isomorphisms

### Definition of an Isomorphism

A linear transformation  $T: V \rightarrow W$  is an **isomorphism** if it is both injective (one-to-one) and surjective (onto).

- **Injective (One-to-One):**  $T(\mathbf{u}) = T(\mathbf{v})$  implies  $\mathbf{u} = \mathbf{v}$ . Equivalently,  $\ker(T) = \{\mathbf{0}\}$ .
- **Surjective (Onto):** For every  $\mathbf{w} \in W$ , there exists a vector  $\mathbf{v} \in V$  such that  $T(\mathbf{v}) = \mathbf{w}$ . Equivalently,  $\text{range}(T) = W$ .

### Isomorphic Vector Spaces

Two vector spaces  $V$  and  $W$  are **isomorphic** if there exists an isomorphism  $T: V \rightarrow W$ . This means they are essentially the same, differing only in notation.

### Theorem

If  $T: V \rightarrow W$  is an isomorphism, then the inverse transformation  $T^{-1}: W \rightarrow V$  is also a linear transformation (and therefore also an isomorphism).

### Theorem

Two finite-dimensional vector spaces  $V$  and  $W$  are isomorphic if and only if they have the same dimension:  $\dim(V) = \dim(W)$ .

### Standard Isomorphism

Every  $n$ -dimensional vector space  $V$  is isomorphic to  $\mathbb{R}^n$ . A standard isomorphism  $T: V \rightarrow \mathbb{R}^n$  is defined by

$$T(\mathbf{v}) = [\mathbf{v}]_B,$$

where  $B$  is any ordered basis for  $V$ .

### Isomorphic Linear Transformation

- Let  $T$  be a linear transformation from  $V$  to  $W$ .
- If  $T$  is onto, then  $\text{rank}(T) = \dim(W)$
- If  $T$  is one-to-one, then  $\text{nullity}(T) = 0$
- The kernel contains more than the zero vector only if the transformation is not one-to-one.
- $T$  is an isomorphism if both of the above is correct